UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. A storyboard is a visual plan of a multimedia sequence.
2. Interlacing displays every line of a video frame at the same time.
3. Frame rate is the number of video frames displayed per second.
4. Streaming allows media to begin playing before the whole file is downloaded.
5. Path animation moves an object along a predefined route.
6. RAM is a type of permanent storage.
7. A scanner is an input device.
8. Sans serif fonts are generally preferred for screen display.
9. Kerning adjusts the space between individual character pairs.
10. A prototype is an early model of a product.
11. A multimedia project always uses every media element.
12. A hyperlink can link only to another web page, not to a file.
13. Authoring tools help developers organise and edit multimedia elements.
14. MIDI files store digitized audio samples.
15. A higher sampling rate generally produces a more accurate recording.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A. Matching options

A. Sampling	B. Frame rate	C. Storyboard
D. Tweening	E. Scanner	F. Card-based tool
G. Chroma key	H. Hypermedia	I. Multimedia
J. Browser	K. Cel animation	L. Virtual reality
M. Keyframe	N. Plotter	O. Linear multimedia

1. integration of text, graphics, audio, video and animation
2. simulated environment
3. important frame in an animation
4. software that displays web pages
5. presentation without user control of sequence
6. frames displayed per second
7. replacing a colour background
8. generating in-between frames
9. converting sound into digital values
10. visual sequence plan
11. linked multimedia elements
12. input device that digitises documents
13. output device for large drawings
14. authoring tool organised like pages or cards
15. hand-drawn frame-by-frame animation

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PART C (30 MARKS)

Choose the correct answer for each question.

1. A hyperlink links to:
   A. Only audio
   B. Only text
   C. Only images
   D. Another resource

2. A licence gives:
   A. Unlimited bandwidth
   B. Free hardware
   C. Permission to use a work
   D. Ownership of all works

3. Which format is commonly used for bitmap images?
   A. MPEG
   B. HTML
   C. PNG
   D. MIDI

4. A bitmap image is made of:
   A. Pixels
   B. Mathematical objects
   C. Sound samples
   D. Text characters

5. A graphics tablet is used for:
   A. Drawing input
   B. Sound output
   C. Web browsing
   D. Printing

6. RAM is:
   A. An input device
   B. Temporary working memory
   C. Permanent storage
   D. A web browser

7. A browser is used to:
   A. Scan images
   B. Display web pages
   C. Print documents only
   D. Edit audio

8. Which image type scales without losing quality?
   A. Scan
   B. Photograph
   C. Bitmap
   D. Vector

9. Anti-aliasing is used to:
   A. Smooth jagged edges
   B. Increase file size only
   C. Add sound
   D. Compress video

10. A wireframe is:
    A. A sound file
    B. A codec
    C. A low-fidelity layout sketch
    D. A final video

11. CMYK is used mainly for:
    A. Sound mixing
    B. Printed output
    C. Screen display
    D. Animation

12. Resolution refers to:
    A. Number of pixels in an image
    B. Colour name
    C. File type
    D. Font size

13. Which is a 3D animation task?
    A. Sampling
    B. Modelling and rendering
    C. Bookmarking
    D. Kerning

14. Sampling converts:
A. Images to vectors
B. Analog sound to digital values
C. Text to speech
D. Digital video to text

15. MIDI stores:
A. Images
B. Digitized audio samples
C. Musical notes and instructions
D. Video frames

16. Which colour model is subtractive?
A. HSL
B. RGB
C. HSV
D. CMYK

17. A presentation where the user controls the sequence is called:
A. Non-linear
B. Linear
C. Printed
D. Static

18. Cel animation is created by:
A. Drawing each frame by hand
B. Printing images
C. Recording sound
D. Scanning text

19. Hertz measures:
A. Loudness
B. Frequency
C. File size
D. Resolution

20. Kerning adjusts:
A. Space between individual character pairs
B. Font colour
C. Page size
D. Line length

21. Hypermedia is best described as:
A. A single video file
B. Multimedia with hyperlinks
C. Text-only media
D. A printed manual

22. Interactivity in multimedia means:
A. Only text is shown
B. There is no navigation
C. The system plays automatically
D. The user can control content and flow

23. Convergence in multimedia refers to:
A. Merging of media and computing
B. Printing images
C. Recording audio only
D. Deleting media files

24. Which is an example of linear multimedia?
A. An interactive kiosk
B. A game with choices
C. A website with menus
D. A cinema presentation

25. A prototype is:
A. A codec
B. A font
C. An early model of a product
D. A final release

26. An authoring tool is used to:
A. Only browse the web
B. Only record audio
C. Only print text
D. Assemble multimedia elements

27. Chroma key is used to:
A. Scan images
B. Add text
C. Compress audio
D. Replace a colour background

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28. Which is a benefit of multimedia in education?

A. It replaces all teachers
B. It combines visuals and sound
C. It uses only text
D. It removes all interaction

29. Keyframes are:

A. Web links
B. Important frames in an animation
C. Audio samples
D. Colour profiles

30. RGB is used mainly for:

A. Audio recording
B. Video editing only
C. Screen display
D. Printed output

PART D (40 MARKS)

QUESTION 1

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(4 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(4 marks)

QUESTION 2

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

(4 marks)

QUESTION 3

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

(3 marks)

b) Describe TWO factors that affect image quality.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

(4 marks)

QUESTION 4

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(5 marks)

END OF QUESTION PAPER